6th

Friday, September 4, 2026 9:52 AM

Neumann BCs

$-(au')' = f$ in $(0,1)$

$u(0) = 0$, $a(1)u' = g_1$

$u(0) = 0$: Dirichlet; $a(1)u' = g_1$: Neumann

$$V = \left\{ v : \int [v^2 + (v')^2]\, dx < \infty , \underbrace{v(0) = 0}_{\text{strong enforced}} \right\}$$

FEM Method:

$u(x) = \sum_{j=1}^{M+1} \xi_j \phi_j(x)$ → trial solution, $+1$ → because last node solution is unknown

$$\int_0^1 a u' v' dx = \int_0^1 f v\, dx + \underbrace{g_1 v(1)}_{\text{Neumann - weakly enforced}}$$

$v = \phi_i$; $i = 1, 2, \dots, M+1$

ex $a = 1$, $h_i = h$ (uniform mesh)

$$\sum_{j=1}^{M+1} \xi_j \int_0^1 \phi_i' \phi_j' dx = \int_0^1 f \phi_i dx + g_1 \phi_i(x)$$

$$A(m, M+1) = \int_{x_m}^{x_{m+1}} \phi_{m+1}' \phi_m' dx$$

$$A(m, m+1) = \int_{x_m}^{x_{m+1}} \phi_m \, \phi_{m+1} \, dx$$

$$= -\frac{1}{h^2} \int_{x_m}^{x_{m+1}} dx$$

$$= -\frac{1}{h}$$

So, $A(m+1, m) = -\frac{1}{h}$ → symmetric matrix

$$A_{m+1,m+1} = \int_{x_m}^{x_{m+1}} (\phi'_{m+1})^2 \, dx = \frac{1}{h}$$

So, $\underline{\underline{A}} \, \underline{\xi} = \underline{b}$ where, $\underline{\xi} = (\underline{\xi}_1, \underline{\xi}_2, \underline{\xi}_3 \cdots \underline{\xi}_{m+1})$

Here, $A = \frac{1}{h}$ [2 −1 ; −1 2 −1 ; ⋱ ⋱ ⋱ ; −1 2 −1 ; −1 1]

$A \in R^{(m+1)\times(m+1)}$, symmetric, positive

$\underline{\underline{A}} \in R^{(m+1)\times(m+1)}$, symmetric, positive definite

$$b_i = \int_{x_{i-1}}^{x_{i+1}} f \phi_i \, dx \quad \text{when,} \quad i = 1, \ldots m$$

$$b_{m+1} = \int_{x_m}^{x_{m+1}} f \phi_{m+1} \, dx + \partial_1$$

Note: for 2-p BVP

(1) dirichlet BCs are strongly enforced (essential)

(2) Neumann or mixed BCs are weakly enforced. (natural BCs)

Error Analysis:

Two kinds of error estimate:

(1) a priori error estimation: error estimate in ters of derivative of exact solution. e.g $\|u - U\| \le C h^p \|u^{(p)}\|$

u: exact, U: FEA, C: constant, p: upper B..., p^{th} derivative of u

upper Bound | e^{th} derivative of u

$P \rightarrow$ order of shape function

(2) a ~~post-priori~~ : posteriori

"error estimates in terms of a residual"

(e.g) $\rightarrow \mathcal{L}(U) - f$

useful for calculating error during the simulation

$\Rightarrow$ adaptive error control (adaptive mesh refinement)

let $e(x) := u(x) - U(x)$ be the error.

"Energy" norm : $\|V\|_E = \sqrt{\int_0^1 a(x)\,[V'(x)]^2\,dx}$

weighted L_2-norm : $\|w\|_a = \sqrt{\int_0^1 a(x)\,[w(x)]^2\,dx}$

Note: $\|V\|_E = \|V'\|_a$

Cauchy inequalities:

$$\left| \int_0^1 a v' w' \, dx \right| \leq \|v'\|_a \, \|w'\|_a$$

$$\Downarrow$$

$$\|v\|_E \, \|w\|_E$$

also,

$$\left| \int_0^1 v w \, dx \right| \leq \|v\|_a \, \|w\|_{a^{-1}}$$

An a priori error estimate:

cG(1) for homogeneous Dirichlet BC

$$\begin{cases} -(au')' = f & \text{in } (0,1) \\ u(0) = u(1) = 0 \end{cases}$$

error calculation

$$\int_0^1 a U' v' \, dx = \int_0^1 f v \, dx \rightarrow \forall \quad v \in V_h \subset V$$

Then

$$\int_0^1 a U' v' \, dx = \int_0^1 f v \, dx \rightarrow \forall \quad v \in V$$

Now subtract

$$\int_0^1 a(u-v')\, v'\, dx = 0 \qquad \forall\ v \in V_h$$

↳ Galerkin orthogonality

For Energy norm error estimates a priori error

$$\|u-v\|_E^2 = \|(u-v)'\|_a^2$$

$$= \int_0^1 a(u-v')(u-v)'dx$$

for any $v \in V_h$

$$\|u-v\|_E^2 = \int_0^1 a(u-v)'(u-v+v-v)'dx$$

$$= \int_0^1 a(u-v)'(u-v)'dx + \int_0^1 a(u-v)'(v-v)'dx$$

0 by Galerkin orthogonality

Note:

$u-v \in V$ but $v-v \in V_h$ →

~~Note:~~ So, $\|u-U\|_E^2 = \int_0^1 a(u-U)'(u-v)'\,dx$

for any $v \in V_h$

using cauchy:

$$\int_0^1 a(u-U)'(u-v)'\,dx \le \|(u-U)'\|_a \|v-U)'\|_a$$

$$\|u-U\|_E^2 \le \|u-U)'\|_a \|(u-v)'\|_a$$

$$\Rightarrow \|(u-U)\|_E^2 \le \|(u-U)'\|_E \|u-v\|_E$$

$$\therefore \|u-U\|_E \le \|u-v\|_E \quad \forall\ v \in V_h$$

FE solution U is the best solution means smallest the energy norm.